

Software Quality Assurance

Lecture 7

Outline

- Formal Technical Reviews

– For verification – Deskchecks – Walkthroughs – Inspections

- Software Configuration Management
  - GIT
- Seven Basic Quality Tools

Formal technical reviews

- A Formal Technical Review (FTR) is a Verification activity performed by software engineers.
- The FTR is actually a class of reviews that include **walkthroughs, inspections,** and **desk checks.**
- A review is performed on any “work product”.
- Each FTR is conducted as a meeting.
- It is successful only if it is properly planned, controlled, and attended by those for whom it was intended
- Objective:
  - To point out needed improvements in a product of a single person or team
  - Confirm those parts of a product in which improvement is either not desired or not needed

Walkthroughs

- Walkthroughs are informal meetings but with a purpose.
- Also known as “team debugging” or “peer reviews”
- An in-depth technical review of some aspect of a software system
- Can be used through out the s/w life cycle

Inspection

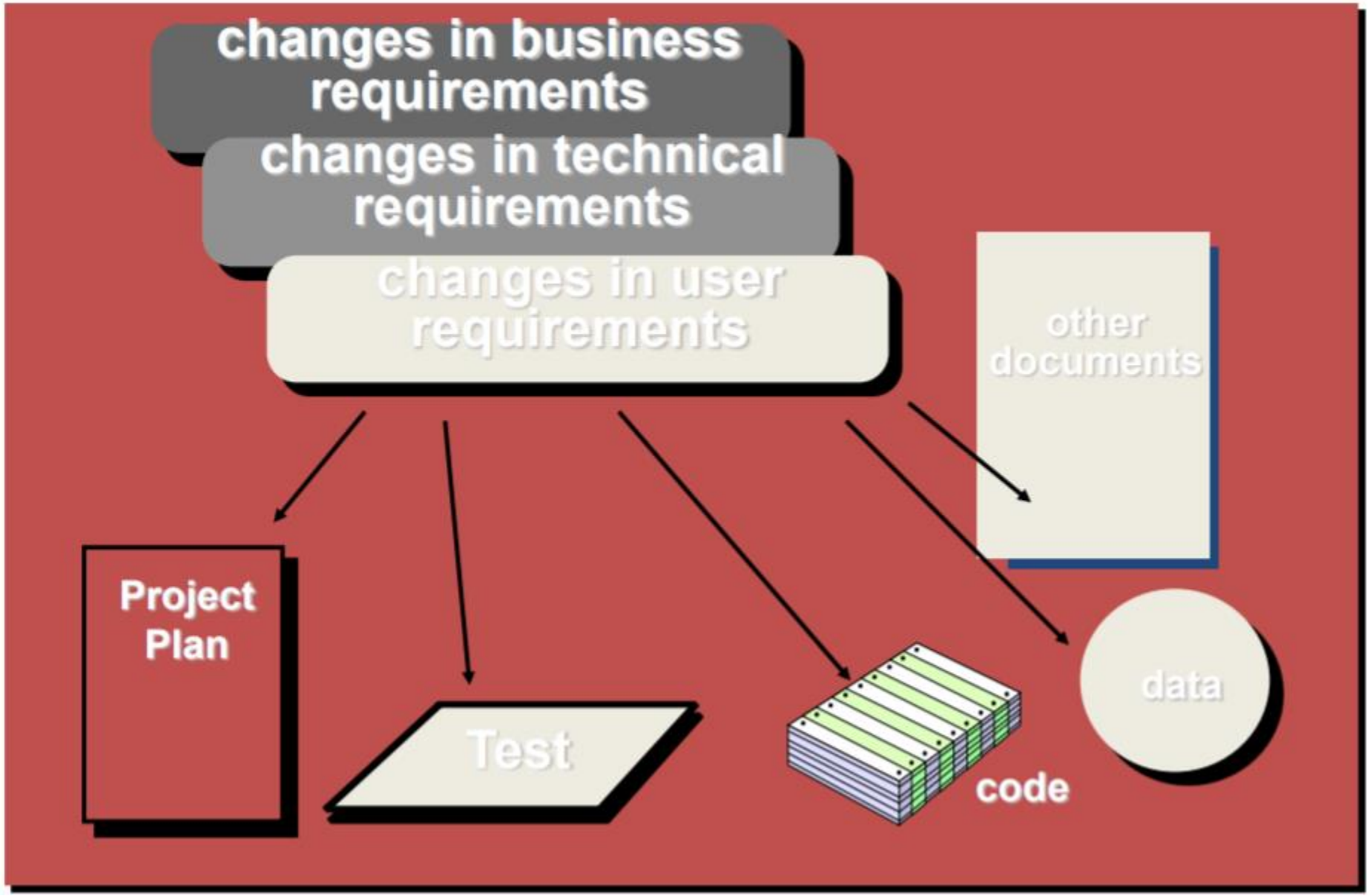
- Inspections are more structured and effective than walkthroughs
- Inspectors have definite role to play
- The moderator rather than the reviewee directs the effort
- An error checklist is used
- Detailed error feedback is provided to individual programmers

Software Configuration Management

- Definition of CM
- What is CM
- Why do we have CM

No matter where you are in the system life cycle, the system will change, and the desire to change it will persist throughout the life cycle.

Bersoff, et al, 1980


changes in business requirements
changes in technical requirements
changes in user requirements
other documents
Project Plan
Test
code
data

Configuration management is the art of identifying, organizing, and controlling modifications to the software (& documentation) being built by a programming team.

SCM activities:

- identify change
- control change
- ensure that change is being properly implemented
- report change to others who may have an interest

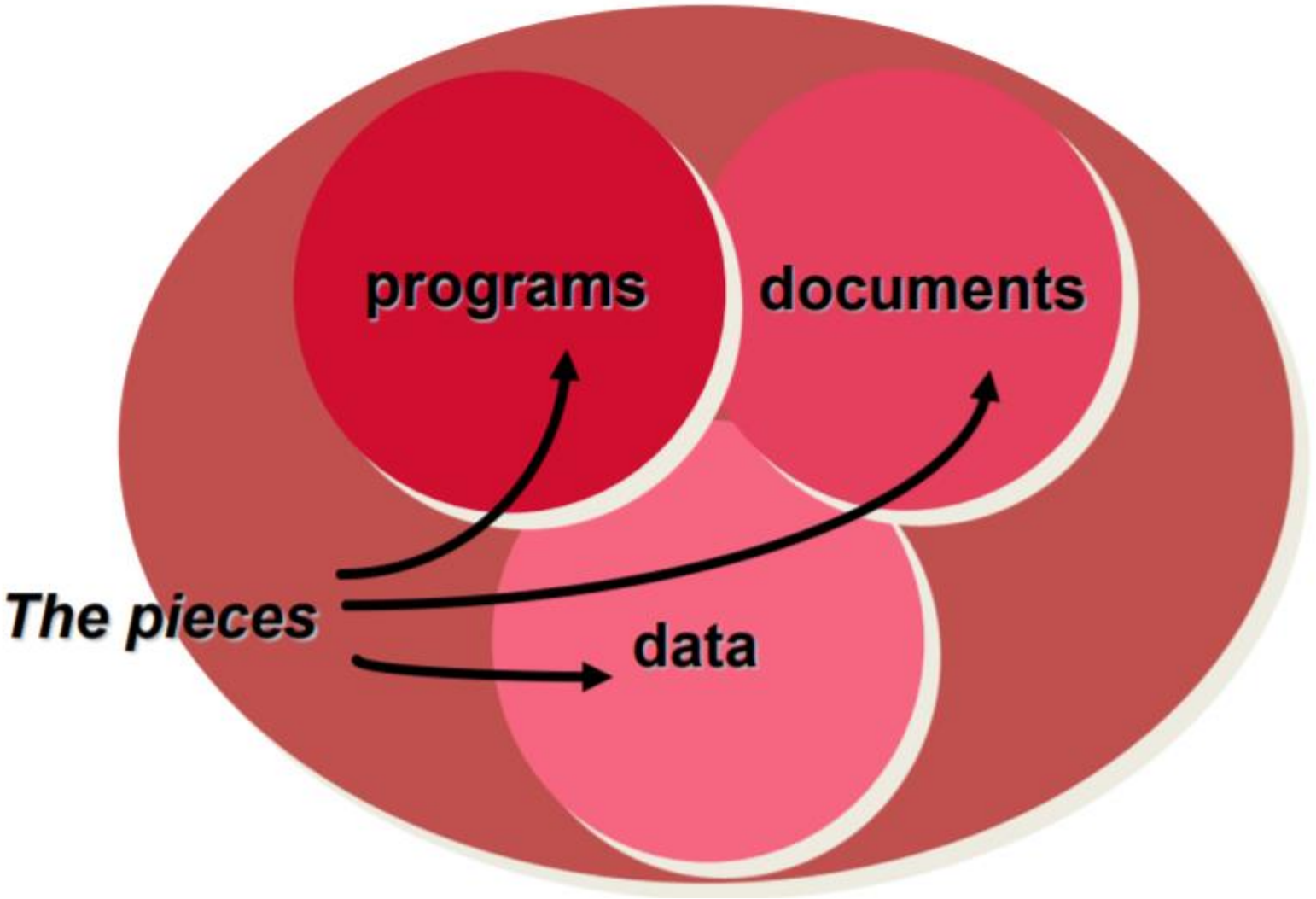
Baselines

"A specification or product that has been formally reviewed and agreed upon, that thereafter serves as the basis for further development, and that can be changed only through formal change control procedures."

- A baseline is a specification or product that has been reviewed and agreed upon
- Once a baseline is established, changes can be made only via a formal procedure, which evaluates and verifies each change.

The Questions Which SCM Must Answer

- How do we manage requests for change?
- What and where are the software components?
- What is the status of each software component?
- How does a change to one component affect others?
- How do we resolve conflicting changes?
- How do we maintain multiple versions?
- How do we keep the system up to date?


programs
documents
The pieces
data

SCM Tasks

- According to IEEE we can define these primary activities in CM
  1. Identification of objects
  2. Management
  3. Configuration auditing
  4. Status reporting

1. Identification of objects

- Identification is the task of identifying our divisional artifacts i.e. the items that make up our system e.g.
  - SDLC Documents
  - Software
  - Hardware

2. Management

- Management is the introduction of controls (procedures and quality gates) to ensure that product evolves appropriately
- Keep your focus on
  - Version control
  - Change control

Change Control Procedure

- Need for change is recognized
- User creates Change Request (CR)
- Developer evaluates CR, and produces report
- Change Control body decides, either
  - Yes: ECO (engineering change order) is queued
  - Maybe: Report rejected – need more info.
  - No: CR is denied; user is informed

3. Configuration Audit

- Audit is the review of the organizational process against the defined/required standards
- Areas of focus includes
  - Adherence to process
  - Conformance to security
  - Configuration Verification

4. Status reporting

- Configuration Status Reporting
- A CSR report is generated on regular basis and is intended to keep management and practitioners appraised of important changes
- What happened?
- Who did it?
- When did it happen?
- What else will be affected?

Cost of quality

Cost of quality (COQ) is defined as a methodology that allows an organization to determine the extent to which its resources are used for activities that prevent poor quality, that appraise the quality of the organization's products or services, and that result from internal and external failures.

Quality-related activities that incur costs may be divided into prevention costs, appraisal costs, and internal and external failure costs

Cost of quality

COQ = Failure Cost, Appraisal Cost & Prevention Cost

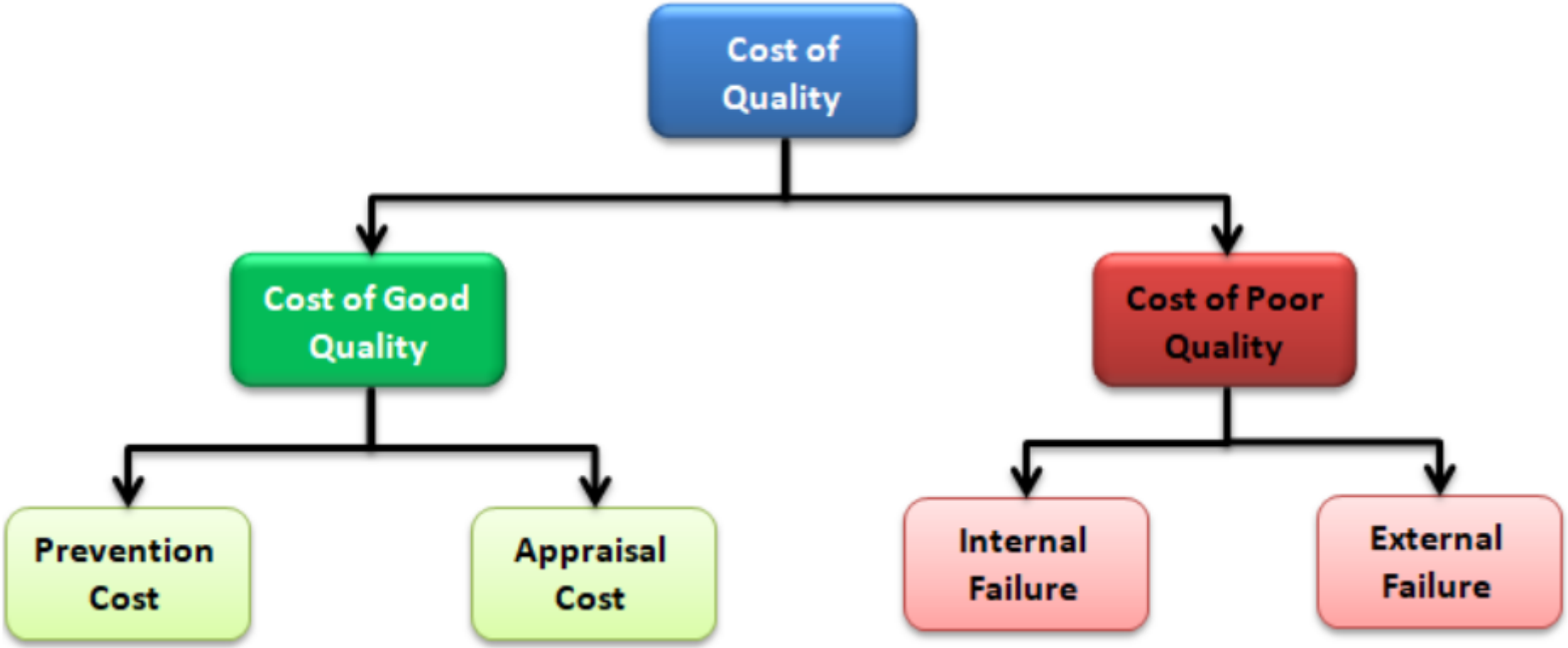
Failure Cost
- Cost of fixing defects
- running debugger to find defects
- repairing time, redesigning, recompiling or retesting

Appraisal Cost
- work to assess the product to see if it has defects
- excluding time spent on fixing
- does not include repair cost
- cost paid to assure that product is defect free

Prevention Cost
- cost of modifying the process
- ex. Analysis done to understand defects
- redesign and test of new process
- building prototype



Cost of Quality Example


Cost of Quality
Cost of Good Quality
Cost of Poor Quality
Prevention Cost
Appraisal Cost
Internal Failure
External Failure

Example

- **Cost of quality = CoGQ + CoPQ**
  - **CoGQ cost of good quality**
  - **CoPQ cost of poor quality**

Simplified Calculations

- Appraisal COQ.. Sum of all review time as % of Total Development Time
- Failure COQ.. Sum of all compile + Test time as % of Total Development Time

Seven Basic Quality Tools

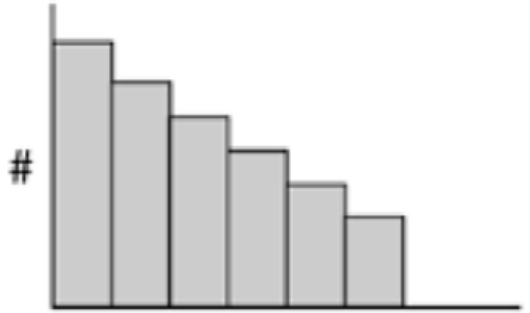
- The basic statistical tools for quality control promoted by Ishikawa (1989) are widely used in manufacturing productions.
- Known as Ishikawa's seven basic tools.
- These statistical tools are for process and quality control at the project and organization level and, hence, are useful for project leaders and process experts.

- they do not provide specific information to software developers on how to improve the quality of their designs or implementation.
- In software development, however, the process is complex and involves a high degree of creativity and mental activity.
- It is extremely difficult, if not impossible, to define the process capability of software development in statistical terms.
- However, good use of the seven basic tools can lead to positive long-term results for process improvement and quality management in software development

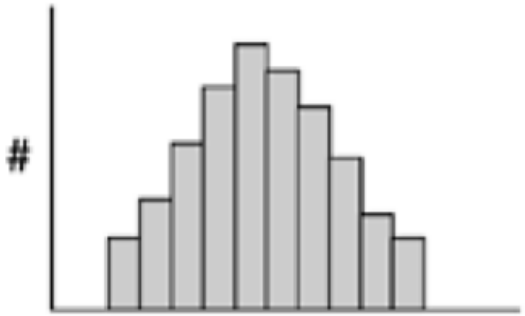
7 basic quality tools

A	II	I	III					
B			I	II	III	I		
C	I	II	II					
D					I	II	II	I

Check Sheet



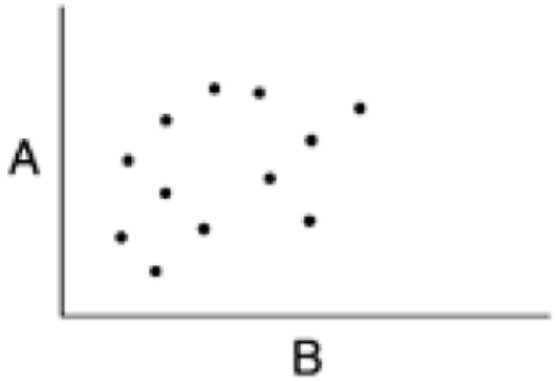
Causes
Pareto Diagram



Units
Histogram



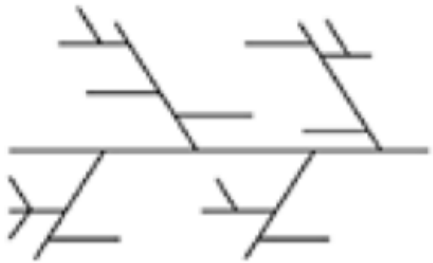
Time
Run Chart



Scatter Diagram



Control Chart



Cause-and-Effect, or Fishbone, Diagram

Check sheet

- A check sheet is a paper form with printed items to be checked.
- Its main purposes are to facilitate gathering data and to arrange data while collecting it so the data can be easily used later.
- Another type of check sheet is the check-up confirmation sheet. It is concerned mainly with the quality characteristics of a process or a product. To distinguish this confirmation check sheet from the ordinary data-gathering check sheet, we use the term checklist

Pareto diagram

- A Pareto diagram is a frequency chart of bars in descending order; the frequency bars are usually associated with types of problems.
- In software development, the X-axis for a Pareto diagram is usually the defect cause and the Y-axis the defect count.
- By arranging the causes based on defect frequency, a Pareto diagram can identify the few causes that account for the majority of defects.
- It indicates which problems should be solved first in eliminating defects and improving the operation.
- Pareto analysis is commonly referred to as the 80–20 principle (20% of the causes account for 80% of the defects), although the cause-defect relationship is not always in an 80–20 distribution

Histogram

- The histogram is a graphic representation of frequency counts of a sample or a population.
- The X-axis lists the unit intervals of a parameter (e.g., severity level of software defects) ranked in ascending order from left to right, and the Y-axis contains the frequency counts.
- In a histogram, the frequency bars are shown by the order of the variable whereas in a Pareto diagram the frequency bars are shown by order of the frequency counts.
- The purpose of the histogram is to show the distribution characteristics of a parameter such as overall shape, central tendency, dispersion, and skewness.
- It enhances understanding of the parameter of interest

Scatter diagram

- A scatter diagram vividly portrays the relationship of two interval variables.
- In a cause-effect relationship, the X-axis is for the independent variable and the Y-axis for the dependent variable.
- Each point in a scatter diagram represents an observation of both the dependent and independent variables.
- Scatter diagrams aid data-based decision making
- (e.g., if action is planned on the X variable and some effect is expected on theY variable).

Run chart

- A run chart tracks the performance of the parameter of interest over time.
- The X-axis is time and the Y-axis is the value of the parameter.
- A run chart is best used for trend analysis, especially if historical data are available for comparisons with the current trend.
- Ishikawa (1989) includes various graphs such as the pie chart, bar graph, compound bar graph, and circle graph under the section that discusses run charts.
- An example of a run chart in software is the weekly number of open problems in the backlog; it shows the development team's workload of software fixes.

Control chart

- A control chart can be regarded as an advanced form of a run chart for situations where the process capability can be defined.
- It consists of a central line, a pair of control limits (and sometimes a pair of warning limits within the control limits), and values of the parameter of interest plotted on the chart, which represent the state of a process.
- The X-axis is real time.
- If all values of the parameter are within the control limits and show no particular tendency, the process is regarded as being in a controlled state.
- If they fall outside the control limits or indicate a trend, the process is considered out of control.
- Such cases call for causal analysis and corrective actions are to be taken.

Fishbone

- The cause-and-effect diagram, also known as the fishbone diagram, was developed by Ishikawa and associates in the early 1950s in Japan.
- It was first used to explain factors that affect the production of steel.
- It shows the relationship between a quality characteristic and factors that affect that characteristic.
- Its layout resembles a fishbone, with the quality characteristic of interest labeled at the fish head, and factors affecting the characteristics placed where the bones are located.
- While the scatter diagram describes a specific bivariate relationship in detail, the cause-and-effect diagram identifies all causal factors of a quality characteristic in one chart.

Checklist

- checklists that summarize the key points of the process are much more effective than the lengthy process documents
- Several examples of checklists are
  - design review checklist
  - code inspection checklist
  - moderator (for design review and code inspection) checklist
  - pre-code-integration (into the system library) checklist
  - entrance and exit criteria for system tests checklist
  - product readiness checklist.

Next agenda

- SW testing